

CAAD-Gate: A Lightweight Context Aware Adaptive Data Gate for Resource Constrained Edge Sensing

Abstract

Edge sensing devices are now used in assistive technology, environmental monitoring, smart homes, safety systems, and Internet of Things applications. These devices often use small microcontrollers such as ESP32 based platforms, which have limited processing power, memory, and battery capacity. A key problem is that sensors continuously collect data, but most of that data may not contain useful events. For example, a sound monitoring device may capture long periods of background noise while only a few short sound events are important. If every frame is sent to feature extraction, recognition, or alert processing, the device wastes processing power and energy.

This study proposes CAAD-Gate, a lightweight Context Aware Adaptive Data Gate for resource constrained edge sensing systems. The purpose of CAAD-Gate is to work as an early filtering layer before heavier processing. Instead of processing every sensor frame, CAAD-Gate first decides whether the current frame is normal, low relevance, or useful enough to send forward. In this way, only meaningful data are passed to later processing stages.

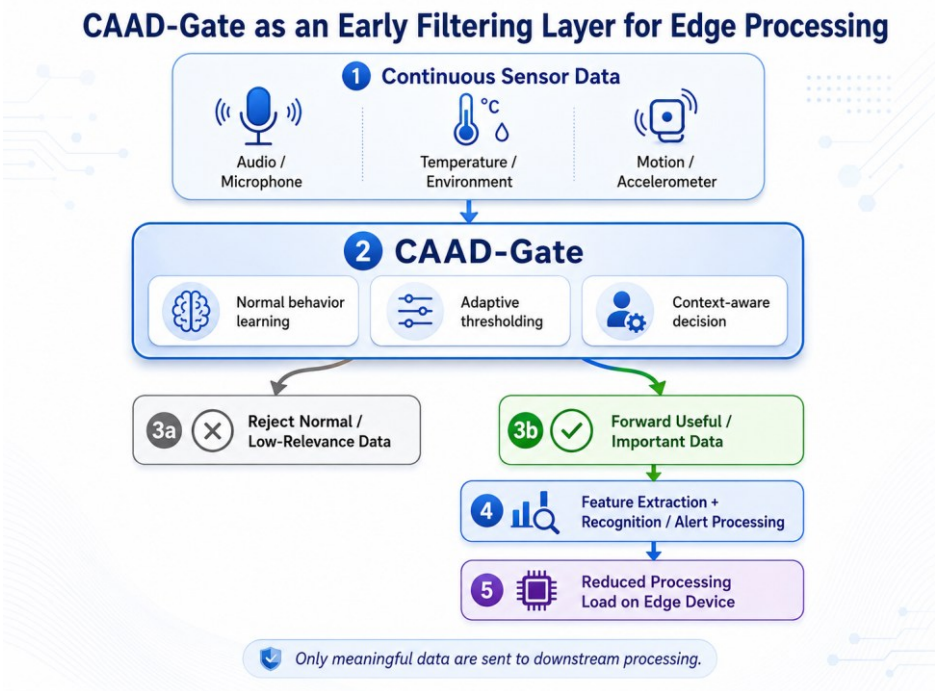


Figure 1. CAAD-Gate as an early filtering layer for reducing unnecessary edge processing.

Environmental sound recognition was used as the first validation case. Audio data were converted into 16 kHz mono signals and divided into 512 sample frames with 50%

overlap. Each frame was processed using DC offset removal and pre emphasis, and then frame energy was calculated. In the previous fixed threshold method, a frame was accepted only when its energy was above one selected global threshold. However, a single fixed threshold may not work well for all sound classes, background levels, and event durations.

To address this limitation, CAAD-Gate uses an adaptive noise floor method. The system estimates the background noise level from early frames and updates it slowly during inactive frames. The active decision is then made using an adaptive threshold based on the current noise floor and a multiplier. Active frames are grouped into sound events using gap frame and minimum event length rules. These events are then evaluated using prototype based recognition with Mel Filter Bank Energies (MFBE) and Mel Frequency Cepstral Coefficients (MFCC).

The experiments used the ESC-50 environmental sound dataset with 2000 WAV recordings. Folds 1 to 4 were used for training with 1600 recordings, and fold 5 was used for testing with 400 recordings. A fixed-threshold baseline was tested using $\text{ENERGY_THRESHOLD} = 40000$, $\text{GAP_FRAMES} = 2$, and $\text{MIN_EVENT_FRAMES} = 6$. This fixed method achieved 93.50% detected-WAV coverage, 1037 test events, and 4011 training prototypes. Its recognition performance was 25.94% MFBE Top-1 accuracy, 46.29% MFBE Top-5 accuracy, 20.44% MFCC Top-1 accuracy, and 42.04% MFCC Top-5 accuracy.

The best adaptive CAAD-Gate setting used an adaptive multiplier of 3.0, a noise update rate of 0.01, 20 noise initialization frames, 4 gap frames, and a minimum event length of 6 frames. It achieved 78.50% detected WAV coverage, 1110 test events, 4417 training prototypes, and 65.33% estimated frame level processing reduction before downstream recognition. Its recognition performance was 24.50% MFBE Top-1 accuracy, 46.04% MFBE Top-5 accuracy, 19.37% MFCC Top-1 accuracy, and 41.08% MFCC Top-5 accuracy.

The results show that CAAD-Gate can reduce unnecessary frame level processing while keeping recognition performance close to the fixed threshold baseline. However, the adaptive gate detected fewer WAV files than the fixed method, showing an important trade off between event coverage and processing reduction. Therefore, CAAD-Gate is a promising lightweight pre processing framework for resource constrained edge sensing. The study also gives a clear experimental foundation for using an adaptive gate before feature extraction or recognition on low power embedded devices, although further tuning is needed to improve coverage while maintaining efficient operation.

Keywords: Adaptive Data Gate; Context Aware Sensing; Edge Intelligence; Environmental Sound Recognition; Resource-Constrained Systems